

EternaCare Documentation-to-Application Alignment Report

Comparison of the previously uploaded project report with the current application

Compared document: EternaCare RSW - Aligned 2026

Drive document last modified: 11 July 2026

Application audit date: 31 July 2026

Audit basis: Current Flutter, Supabase SQL, Edge Function, Python worker, and automated test files in the project workspace

Overall conclusion: the document requires a substantial update.

The central project description remains correct, but development after 11 July introduced important authentication, reward administration, Legacy Checking, scheduled-server, contact, user-interface, and testing changes. The report should not be submitted as a current implementation description until the major findings below are corrected.

1. Scope and Confidence

This comparison checks what is present in the current source code and migration files. It distinguishes implemented repository behavior from deployment-dependent behavior.

- **Verified in code:** Flutter screens and services, validation, local background scheduling, Supabase migrations, Edge Functions, Python worker equivalent, and automated tests.
- **Verified by execution:** 63 Flutter tests passed on 31 July 2026. Android and Windows debug builds completed successfully.
- **Requires live dashboard confirmation:** whether every migration is applied, every Edge Function is the newest deployed version, the Cron job is active, Realtime publication is enabled, and Twilio/Brevo/provider secrets are valid.
- **Not changed by this audit:** the previously uploaded Drive report and the application source code.

2. Findings Overview

No.	Finding	Status	Main sections affected
1	Server-scheduled 90-day Legacy inactivity processing is missing from the report.	MAJOR UPDATE	Abstract, 3.2, 4.1, 4.2, 5, 6, 7
2	The owner warning, 24-hour cancellation period, seven-day release window, and split access scope are undocumented.	MAJOR UPDATE	3.2, 4.2, 4.3, 4.4, 5, Appendix A

No.	Finding	Status	Main sections affected
3	The report says there is no administrator dashboard, but a separate protected reward-admin interface now exists.	MAJOR UPDATE	3.2.1, 3.2.2, 4.1, 4.5, 5, 6
4	Reward behavior now includes virtual badges, virtual vouchers, personal redemption codes, badge collection, refresh, Realtime, and admin deletion rules.	MAJOR UPDATE	3.2.2, 4.3, 4.5, 5, Appendix A/B
5	Authentication now includes forgot-password recovery, eight-digit email codes, duplicate-email handling, and optional biometric session unlock.	MAJOR UPDATE	Abstract, 3.2.2, 4.2.1, 4.4, 5, Appendix A
6	The primary contact now stores an email address for Legacy release notices.	MAJOR UPDATE	3.2.2, 4.3 ERD, 4.5 contact UI, Appendix A
7	The ERD and security discussion omit new Legacy, reward-admin, terms, and redemption attributes/tables.	MAJOR UPDATE	4.3, 4.4
8	The report states that Home shows check-in totals and streak, but these were removed from Home and the Oren area was redesigned.	MEDIUM UPDATE	4.5 UI Prototype, Appendix A
9	The deployment section describes Android and optional web testing, while the current project also has an active Windows desktop build.	MEDIUM UPDATE	Abstract, 5, 6, Appendix B
10	Terms and guidance content now covers biometric unlock, virtual rewards, and the Legacy release policy, and is accessible from Profile.	MEDIUM UPDATE	3.2.2, 4.4, 4.5, Appendix A
11	The document reports 26 passing tests; the current Flutter suite has 63 passing tests.	MAJOR UPDATE	Abstract, Chapter 5, Chapter 7

3. Detailed Differences

3.1 Two Different Background Processes Now Exist

Previously documented: Android WorkManager performs a best-effort check every 15 minutes, and server-scheduled monitoring is listed as future work.

Current application: The WorkManager description remains correct for the phone-side three-reminder safety flow. A second, independent server-side process now handles the 90-day Legacy heartbeat lifecycle. Supabase Cron runs at `0 16 * * *`, which is 12:00 AM Malaysia time, and invokes the `process-legacy-inactivity` Edge Function. A FastAPI/Python worker exists as an equivalent external-server implementation, but deployment documentation says to run either the Python target or the TypeScript Edge Function target, not both.

Important rule: Only a successful row in `public.checkins` is a Legacy heartbeat. SOS and SMS alarm records do not reset the 90-day clock.

Required correction: Keep the existing WorkManager explanation, add a separate server-scheduled Legacy monitoring subsection, and remove "server-scheduled monitoring" from future work.

3.2 Legacy Checking Has a Protected Release Lifecycle

Previously documented: Legacy Planning is mainly described as authenticated CRUD for funeral preferences, notes, and private documents.

Current application:

1. The owner must enable Legacy Checking.
2. The system records consecutive no-heartbeat days in a server-owned status table.
3. At day 90, the system emails the account owner first.
4. The owner has 24 hours to check in or use a one-time cancellation link.
5. If there is no cancellation and no newer check-in, the primary contact receives an email.
6. Protected Legacy Notes and signed secure-document links are available for seven days.
7. A new check-in, cancellation, changed eligibility, or expiry revokes protected access.
8. Funeral preferences use a separate scope: an SMS-verified primary contact may view them whenever the owner has enabled Legacy Checking.
9. Legacy UID, matching primary phone, and a six-digit SMS code are still required outside the owner-approved debug testing path.

Required correction: Add activity and sequence diagrams for owner warning, cancellation, primary-contact notification, SMS verification, preference-only access, protected access, expiry, and revocation.

3.3 Authentication and Account Security Expanded

Previously documented: Email/password, generic OAuth, profile bootstrap, manual signup-code verification, and Terms acceptance.

Current application adds:

- Eight-digit signup email verification codes.
- Forgot-password email requests, eight-digit recovery codes, recovery links, and a reset-password screen.

- Duplicate-email detection that directs the user to sign in or use Forgot password.
- Google, Facebook, and GitHub OAuth buttons, subject to provider configuration.
- Optional per-device biometric unlock for a saved Supabase session: Face ID, face recognition, Touch ID, fingerprint, or Windows Hello.
- Automatic biometric re-lock when the app returns from the background.
- Password sign-in remains necessary after explicit sign-out or session expiry.

Clarification: Phone OTP and Legacy Checking SMS OTP remain six digits. Email signup and password-recovery codes are eight digits.

3.4 Contact Data Now Includes Email

Previously documented: Contact name, relationship, verified phone, address, state, region, and primary status.

Current application: The contact model and UI also include a validated email address. The Legacy server uses the primary contact's email for the protected-access notification, while the verified phone remains necessary for identity checks and SMS OTP.

Required correction: Add email to the Contact entity, form description, functional requirements, validation section, and Legacy release sequence.

3.5 Rewards Are Virtual and Separately Administered

Previously documented: The report mentions virtual voucher-style rewards and says there is no administrator dashboard.

Current application:

- All catalog rewards are virtual badges or virtual vouchers; there is no delivery or fulfillment queue.
- Vouchers receive a unique user-specific redemption code generated by Supabase.
- An earned badge is stored in "My Badge List" after the user first selects Collect Badge.
- The reward detail page shows description, voucher value, and redemption code when applicable.
- A separate direct-entry admin interface can create, edit, activate, deactivate, refresh, and select rewards for deletion.
- The normal user navigation does not link to the admin interface.
- Only allowlisted rows in reward_admins may use protected catalog RPCs.
- An earned reward cannot be permanently deleted; it must be deactivated to preserve history.
- Catalog Realtime refresh is implemented, with manual refresh as a fallback.

Required correction: Replace the statement "there is currently no administrator dashboard" with a description of the separate protected reward-admin interface.

3.6 Home and Oren UI Description Is Outdated

Previously documented: The Home page shows streak and check-in totals alongside the other dashboard information.

Current application: Home no longer presents those totals. The latest layout places transparent Oren status content above the Oren scene, shows the selected Malaysia region and weather, displays a token balance, embeds the Shop action in the scene, and places Feed fish, Play, and item selection controls below Oren. Safety

Monitor and inactivity reminders are lower on the page. The guidance control explains Oren interaction and token earning.

Required correction: Replace the Home screenshot and Figure 4.4.1 description. Do not claim that Home shows check-in totals or streak.

3.7 ERD and Security Model Need New Entities

The current ERD discussion should be expanded to include, at minimum:

- `legacy_heartbeat_status`: latest heartbeat, consecutive no-heartbeat days, state, and current access window.
- `legacy_access_windows`: owner warning, cancellation deadline, primary-contact notice, availability, expiry, and state.
- `legacy_release_cancel_tokens`: one-time cancellation token hashes and expiry.
- `legacy_access_audit` and `legacy_server_test_events`.
- `reward_admins`, virtual reward catalog administration, claimed badge state, and per-user voucher redemption codes.
- Terms version and acceptance timestamp captured during signup.
- Contact email and phone-verification timestamp.
- Per-device biometric enablement is local configuration, not a biometric template stored in Supabase.

3.8 Testing Evidence Is Outdated

Previously documented: 26 passing tests.

Current application: The Flutter suite reports 63 passing tests. New coverage includes duplicate signup handling, eight-digit email codes, recovery endpoints, Terms capture, biometric availability and lifecycle locking, reward-admin routing, voucher data, badge collection, Legacy preference-only access, protected waiting states, incorrect primary-phone feedback, seven-day windows, and owner-scoped testing.

Required correction: Update every occurrence of "26 passing tests" and extend the Chapter 5 test-case table. Keep provider-dependent SMS, email, OAuth, weather, AI, location, Cron, and Realtime behavior classified as integration testing.

3.9 Platform and Deployment Scope Expanded

Previously documented: Android deployment with optional web testing.

Current application: Android remains the main phone target, and Windows desktop now builds successfully and supports the separate admin route and Windows Hello. Web remains useful for testing and direct admin routing. iOS and macOS biometric configuration exists in source, but those builds were not verified during this audit.

Required correction: Describe Android as the primary safety deployment, Windows as a supported desktop/admin build, and web as a configured test/admin surface. Avoid claiming that all platform-specific builds were integration-tested.

3.10 Terms and In-App Guidance Changed

The current Terms version is 2026-07-31.1, effective 31 July 2026. The Terms now explain biometric unlock, virtual rewards, Legacy preference access, the 90-day protected release, the owner cancellation period, the seven-day window, and provider limitations. Users must review and accept the Terms during account creation, and they can reopen the Terms from Profile. Contextual guidance controls are also present across major pages.

4. Statements That Remain Aligned

The following parts of the uploaded report remain materially consistent with the current source:

- Ethernacare uses Oren as a low-effort daily safety check-in.
- A user can create at most one check-in record per local calendar day.
- Supabase Auth, PostgreSQL, RLS, private Storage, and Edge Functions provide the cloud backend.
- Android WorkManager performs a best-effort 15-minute local inactivity check.
- The local safety flow progresses through three reminders and escalates on the third threshold.
- Primary-contact mode uses automated SMS where the provider and permissions are configured.
- Real emergency messages include a Google Maps location URL when location is available.
- Malaysia's official weather API is preferred, with Open-Meteo as fallback.
- Gemini is invoked server-side and the app has offline guidance fallback.
- Phone and Legacy SMS OTP codes are generated server-side and stored as hashes.
- Legacy documents are private PDF/JPEG/PNG files with a 10 MB limit.
- Ethernacare does not automatically call Malaysia's 999 service.
- SharedPreferences provides user-scoped local caching for Oren and selected app snapshots.

5. Chapter-by-Chapter Update Checklist

Abstract

- Add server-scheduled Legacy monitoring, owner protection, protected contact access, virtual reward administration, password recovery, and biometric unlock.
- Replace the old automated-test count.

Chapter 3 - Methodology and Requirements Analysis

- Add the reward administrator as a separate protected system actor.
- Expand authentication, rewards, contacts, and Legacy Checking requirements.
- State that the phone-side reminder policy and server-side 90-day policy are different processes.

Chapter 4 - System Design

- Update the architecture diagram with Supabase Cron, the Legacy worker, Brevo owner/contact email, and Twilio verification SMS.
- Add Legacy release and biometric-unlock activity diagrams.
- Update the ERD with the tables and attributes listed in Section 3.7.
- Replace Home, Rewards, Contacts, Profile, Legacy Planning, and Legacy Check screenshots.

Chapter 5 - Implementation and Testing

- Describe the current reward and Legacy implementations.
- Update the test count from 26 to 63 and add the newer test groups.
- Keep external-provider tests separate from deterministic automated tests.

Chapter 6 - System Deployment

- Add the Legacy migrations, Cron/Vault setup, Edge Function deployments, Brevo secrets, reward-admin authorization, Realtime setup, and biometric native configuration.
- Add Windows build and direct admin-route instructions.

Chapter 7 - Discussion and Conclusion

- Move server-scheduled Legacy monitoring from future work to completed implementation.
- Retain limitations about OS scheduling, provider availability, quota, billing, and the lack of medical monitoring.

Appendices

- Update the user guide for password recovery, biometric unlock, virtual reward details, Legacy UID, funeral-preference access, and protected Legacy release.
- Update the developer guide for Cron, Edge Functions, secrets, migrations, reward admin, Realtime, Android biometrics, iOS Face ID text, and Windows Hello.

6. Ready-to-Use Replacement Paragraphs

6.1 Revised Authentication Paragraph

EternaCare supports validated email/password registration and configured OAuth providers. Email signup and password recovery use eight-digit email codes, while profile-phone, contact-phone, and Legacy Checking verification use separate six-digit SMS OTP flows. Duplicate email registration is rejected with guidance to sign in or recover the password. After a Supabase session is established, users may optionally enable per-device biometric unlock using Face ID, face recognition, Touch ID, fingerprint, or Windows Hello. Biometric data remains under operating-system control and is not stored by EternaCare.

6.2 Revised Background Processing Paragraph

EternaCare uses two background mechanisms for different safety responsibilities. Android WorkManager performs best-effort 15-minute checks for the configurable three-reminder inactivity flow on the user's phone. Separately, Supabase Cron runs once daily at 12:00 AM Malaysia time and invokes the Legacy inactivity processor. The server processor derives consecutive inactivity from the latest successful check-in; SOS and SMS alert records do not count as heartbeats. A Python FastAPI worker is maintained as an equivalent deployable implementation, although only one scheduled worker target should be configured in production.

6.3 Revised Legacy Checking Paragraph

Legacy Checking separates funeral-preference access from protected Legacy Notes and secure documents. When the owner enables Legacy Checking, an SMS-verified primary contact can verify the owner's Legacy UID and phone relationship to view funeral preferences. Protected notes and signed document links require a server-created release window after 90 days without a successful check-in. At the threshold, EternaCare emails the account owner and provides a 24-hour opportunity to check in or cancel using a one-time secure token. If the release is not cancelled and no new check-in occurs, the primary contact is notified and protected access opens for seven days before expiring.

6.4 Revised Rewards and Administration Paragraph

EternaCare rewards are virtual badges and virtual vouchers rather than physical products. Users unlock rewards from check-in streaks, collect earned badges into a badge list, and view personal voucher redemption codes in reward details. A separate direct-entry administrator interface manages the virtual catalog through protected Supabase RPCs. Allowlisted administrators can create, edit, activate, deactivate, refresh, and delete eligible catalog records; rewards already earned by users cannot be deleted and must instead be deactivated.

6.5 Revised Testing Paragraph

The current Flutter automated suite contains 63 passing tests. Coverage includes authentication validation, duplicate signup handling, eight-digit email verification and recovery, Terms capture, biometric locking, responsive onboarding, Oren interactions, check-in and inactivity boundaries, contact validation, reward caching, virtual vouchers, reward-admin routing, weather mapping, secure Legacy models, preference-only Legacy access, protected release states, seven-day access parsing, and AI fallback. OAuth, SMS, transactional email, location, weather, AI, Realtime, Cron, and operating-system background execution still require deployed-environment integration testing.

7. Live Deployment Checks Before Final Submission

The report should describe these items as deployed only after they are confirmed in the live Supabase project:

1. All migrations through `202607270002_reschedule_legacy_inactivity_midnight.sql` are applied.
2. The active Cron job is named `ethernacare-legacy-heartbeat-midnight-myt` and uses `0 16 * * *`.
3. `process-legacy-inactivity`, `cancel-legacy-release`, `request-legacy-access`, and `verify-legacy-access` are deployed from the newest source.
4. Brevo sender identity and API secrets are valid for owner and primary-contact emails.
5. Twilio secrets, sender capability, Malaysia Geo Permissions, and trial-recipient restrictions are satisfied.
6. The primary contact has a valid email and SMS-verified phone.
7. Reward catalog Realtime publication is active, and protected reward-admin RPCs are installed.
8. The selected admin account has an active `reward_admins` row.
9. Android, Windows, and provider-dependent flows are retested after the final release build.

8. Repository Evidence Reviewed

Representative source paths used for this comparison:

- `lib/presentation/screen/auth/`, `lib/services/biometric_auth_service.dart`, and `lib/presentation/screen/profile/profile_screen.dart`
- `lib/presentation/screen/home/`, `lib/presentation/screen/rewards/`, and `lib/presentation/screen/admin/`
- `lib/presentation/screen/planning/legacy_planning_screen.dart` and `legacy_check_screen.dart`
- `supabase/migrations/202607220001_legacy_heartbeat_windows.sql`
- `supabase/migrations/202607250001_owner_legacy_release_grace.sql`
- `supabase/migrations/202607250002_legacy_preference_access.sql`
- `supabase/migrations/202607260002_virtual_rewards_only.sql` through `202607270001_claim_badges_on_first_open.sql`
- `supabase/functions/process-legacy-inactivity/`, `request-legacy-access/`, and `verify-legacy-access/`
- `server/python_legacy_server/` and `supabase/LEGACY_INACTIVITY_DEPLOYMENT.md`
- `test/auth_repository_test.dart`, `test/biometric_auth_service_test.dart`, and `test/widget_test.dart`

Final recommendation: update the original report using this document as a change checklist, then replace the UI screenshots and regenerate the Table of Contents, List of Figures, ERD, activity diagrams, and test-case totals.